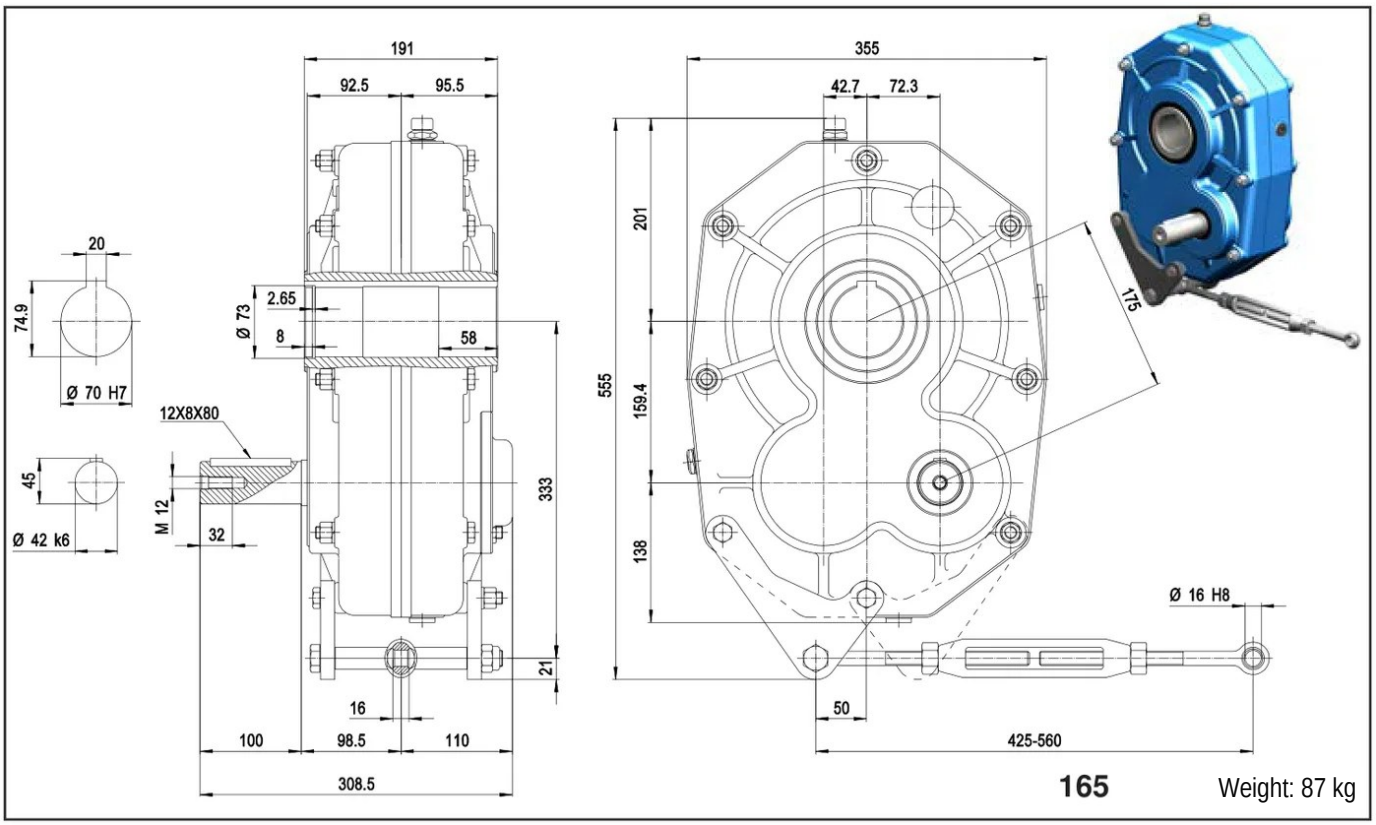
PRODUCT DATA SHEET

# OCR RP 165

Shaft-mounted gear reducer | supplied and supported by Q² Machines

| WEIGHT | RATED POWER | OUTPUT TORQUE | RATIOS |
|---|---|---|---|
| **87 kg** | **4.21–44.2 kW** | **1189–3258 Nm** | **5** |
| complete unit | across listed ratios | catalogue, service factor 1 | single and double stage |

## DIMENSIONS (mm)




## CONSTRUCTION

- Grey cast-iron housing with reinforcing and cooling ribs
- Case-hardened helical steel gears, controlled or ground tooth profile
- Ball and/or taper-roller bearings
- Hollow output shaft in Fe510 steel, with keyway and circlip groove
- Oil-bath lubrication with filler, level and drain plugs
- Double-lip seals, blue RAL 5010 finish
- Torque arm with brackets; backstop available on request

| STAGE | RATIO | OUTPUT n2 (rpm) | INPUT n1 (rpm) | POWER (kW) | TORQUE (Nm) |
|---|---|---|---|---|---|
| Single | 3.21 | 355 | 1140 | 44.2 | 1189 |
| Single | 3.21 | 280 | 899 | 38 | 1295 |
| Single | 3.21 | 220 | 706 | 32.3 | 1401 |
| Single | 3.21 | 180 | 578 | 28.4 | 1507 |
| Single | 3.21 | 140 | 449 | 23.6 | 1613 |
| Single | 3.21 | 110 | 353 | 19.8 | 1719 |
| Single | 3.21 | 90 | 289 | 17.2 | 1826 |
| Double | 11.6 | 110 | 1276 | 20.5 | 1778 |
| Double | 11.6 | 90 | 1044 | 17.8 | 1890 |
| Double | 11.6 | 71 | 824 | 14.9 | 2003 |
| Double | 11.6 | 56 | 650 | 12.4 | 2115 |
| Double | 11.6 | 45 | 522 | 10.5 | 2228 |
| Double | 11.6 | 35.5 | 412 | 8.7 | 2340 |
| Double | 11.6 | 28 | 325 | 7.19 | 2453 |
| Double | 15.1 | 90 | 1359 | 20.8 | 2205 |
| Double | 15.1 | 71 | 1072 | 17.7 | 2381 |
| Double | 15.1 | 56 | 846 | 15 | 2556 |
| Double | 15.1 | 45 | 680 | 12.9 | 2732 |
| Double | 15.1 | 35.5 | 536 | 10.8 | 2907 |
| Double | 15.1 | 28 | 423 | 9 | 3083 |
| Double | 15.1 | 22 | 332 | 7.5 | 3258 |
| Double | 20 | 71 | 1420 | 16.7 | 2250 |
| Double | 20 | 56 | 1120 | 13.9 | 2376 |
| Double | 20 | 45 | 900 | 11.8 | 2502 |
| Double | 20 | 35.5 | 710 | 9.8 | 2628 |
| Double | 20 | 28 | 560 | 8.1 | 2754 |
| Double | 20 | 22 | 440 | 6.64 | 2880 |
| Double | 20 | 18 | 360 | 5.67 | 3006 |
| Double | 22.5 | 56 | 1260 | 13.2 | 2250 |
| Double | 22.5 | 45 | 1013 | 11.1 | 2354 |
| Double | 22.5 | 35.5 | 799 | 9.1 | 2457 |
| Double | 22.5 | 28 | 630 | 7.5 | 2561 |
| Double | 22.5 | 22 | 495 | 6.14 | 2664 |
| Double | 22.5 | 18 | 405 | 5.22 | 2768 |
| Double | 22.5 | 14 | 315 | 4.21 | 2871 |

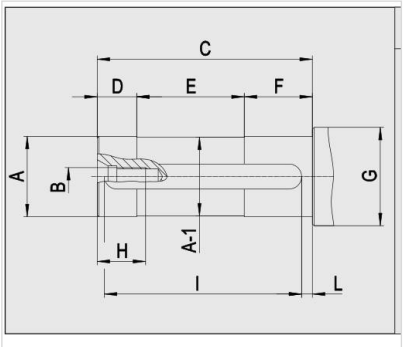
## NOTES

- Rated power and torque are catalogue values for a service factor of 1 (OCR RP 2C catalogue, 2014).
- Maximum input speed 1400 rpm; preferred 710 rpm.
- Efficiency about 0.98 single stage, 0.96 double stage.
- Gear capacity calculated to ISO 6336, DIN 3990 and AGMA 2001-B88.
- Drawings in DXF, PDF and STEP formats, spares, and selection help are available from Q² Machines on request.

## MOUNTING OPTIONS

Shaft-mounted: no coupling, foot or base plate needed

### Installation on the machine shaft

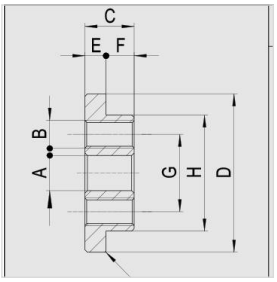



The reducer fits directly onto the shaft of the driven machine. Machine shaft tolerance h6, j6 or k6 as required; bore tolerance H7.

- Mount, fix and remove with tension rods and extractors, using the tapped hole in the end of the machine shaft and an installation washer.
- Clean and lubricate the contact surfaces before mounting to prevent rust.
- Do not force the reducer on with blows or jerks: this damages bearings and seals.
- Mount so the reducer is not subject to vibration.

### Machine shaft, RP 165

| ØA (mm) | B | C | D | E | F | ØG | H | I | L |
|---|---|---|---|---|---|---|---|---|---|
| 70 h6/j6/k6 | M16 | 169 | 36 | 75 | 58 | 80 | 40 | 150 | 9.5 |

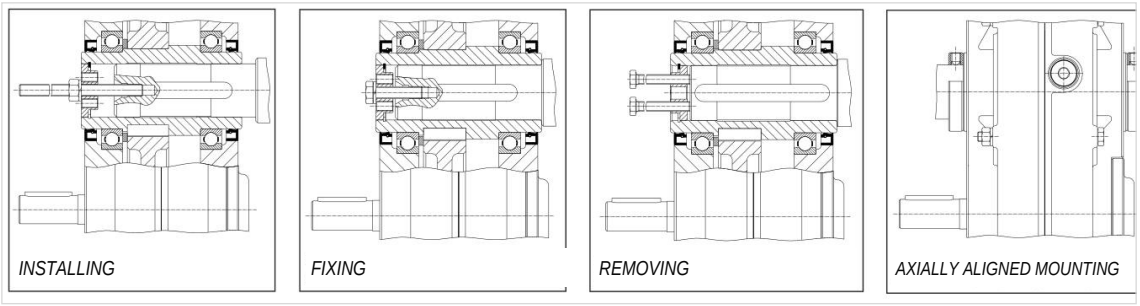



### Installation washer, RP 165

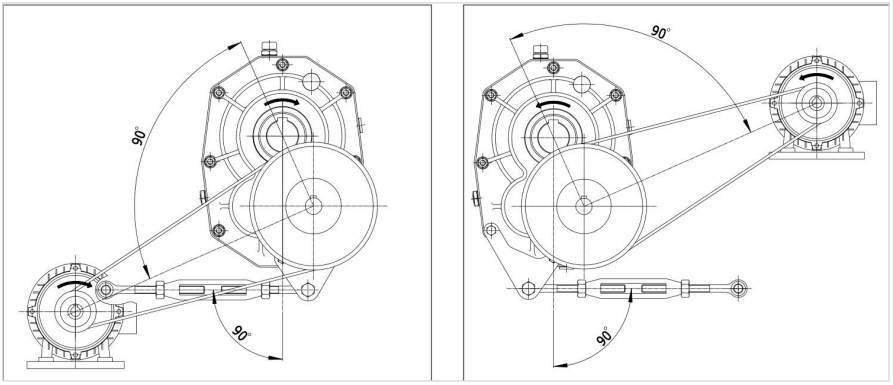
| ØD h10 | ØA | B | C | E | F | G | ØH |
|---|---|---|---|---|---|---|---|
| 70 | 17 (M16) | M12 | 18.5 | 8 | 10.5 | 36 | 54.5 |

Make the washer from the dimensions shown; sharp edge on the outer rim. Bore tolerance H7.

### Installing, fixing, removing and axially aligned mounting



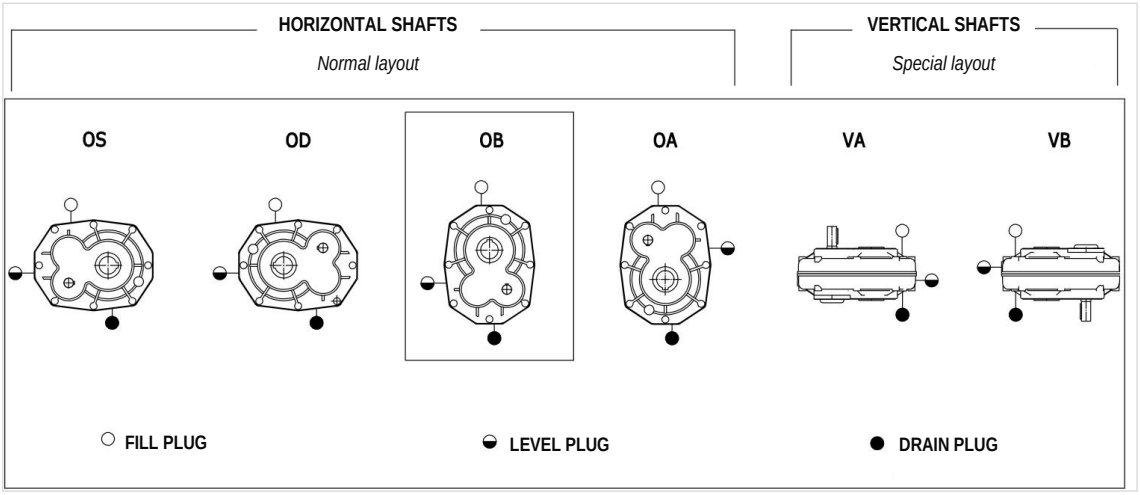

### Torque arm arrangement




- Position the torque arm so it works in tension, at the angles shown. The arm also holds the belts at the right tension.
- For rotation in both directions or heavy duty, use two opposing torque arms.
- Fit torque limiters or safety devices where overloads are foreseen.

## MOUNTING POSITIONS

Standard layout is OB; change position by swapping the plugs




Unless ordered otherwise, reducers are supplied in the OB layout. The other horizontal positions (OS, OD, OA) can be set by the customer by inverting the plugs. Vertical positions VA and VB are special and must be specified with the order.

## LUBRICATION RECOMMENDATIONS

Supplied without oil

RP Series reducers are oil bath and splash lubricated and are supplied WITHOUT OIL. Use ISO VG mineral oil according to the ambient temperature. Fill to the level plug and avoid overfilling. When input speed is below 355 rpm, or in vertical positions, the upper bearings are permanently greased.

| AMBIENT TEMPERATURE | VISCOSITY ISO VG |
|---|---|
| Below -10 °C | **100** |
| -10 to +10 °C | **150** |
| +10 to +30 °C | **220** |
| Above +30 °C | **320** |

### EXAMPLES OF SUITABLE OILS

AGIP Blasia
ARAL Degol BG
BP Energol GR-XP
CASTROL Alpha SP
ESSO Spartan EP
MOBIL Mobilgear Serie 6
SHELL Omala Oil
TEXACO Meropa
TOTAL Carter EP

### Oil quantity for RP 165, litres, double reduction (RP2C)

| OS | OD | OB (standard) | OA | VA | VB |
|---|---|---|---|---|---|
| 4.2 | 3.8 | 3.2 | 6.5 | 3.1 | 5.6 |

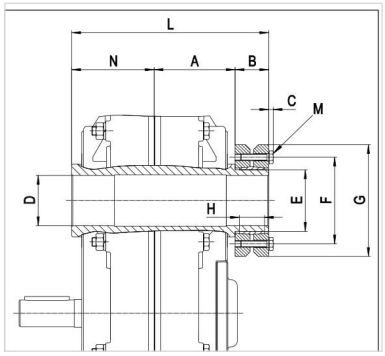
- For positions between those in the table, interpolate. For single reduction units (RP1C, ratio 4), add 10%.
- Change the oil after a running-in period of 500 operating hours, check the level every 1000 hours and change the oil every 4000 hours.
- If oil temperature exceeds 80 °C, or under overloads, change the oil more often.

## HANDLING, STORAGE AND SAFETY

- Allow space around the reducer for inspection and maintenance and enough air flow for cooling. Take care when lifting.
- Protect from weather and direct sun where possible. Store in a dry, clean place; every six months turn the gears a few times and check the grease in the seals and the protective coating on machined parts. Check more often in damp, corrosive or hot-cold environments.
- Fit protection where an accidental oil leak could cause serious damage, and guards over the belt drive.
- Supplied painted blue RAL 5010; shaft ends and seals are protected with water-repellent grease.

## OPTIONS

Available on request; confirm availability with Q² Machines



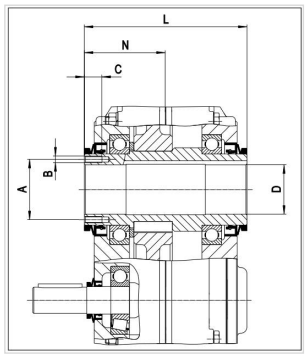

### Shrink disc (keyless locking)

A keyless device to mount and fix the machine shaft to the reducer, removing the key and installation washer and giving positive control of the torque transmitted. A special hollow shaft is needed, with the dimensions below.

**RP 165, mm unless stated**

| ØD H7 | L | N | A | B | C | Screws | ØE h8 |
|---|---|---|---|---|---|---|---|
| 70 | 238 | 104.5 | 101.5 | 32 | 5.5 | M8 | 80 |

| F | ØG | H | Type | Screw torque Nm | Torque Nm | Axial force N | kg |
|---|---|---|---|---|---|---|---|
| 100 | 145 | 25 | d80 | 30 | 4,600 | 131,500 | 1.9 |

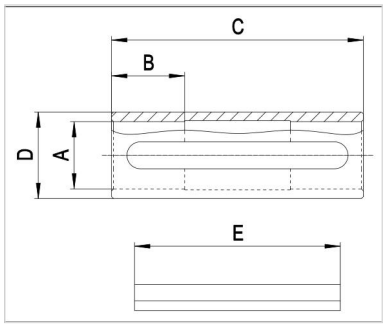



### Special seals

For dusty or aggressive environments where regular seal maintenance is impractical. An additional rotating front seal (V-ring on a wear plate) runs with the shaft against a fixed surface. A special hollow shaft is needed.

**RP 165, mm**

| ØD H7 | L | N | A | B | C | V-ring input | V-ring output |
|---|---|---|---|---|---|---|---|
| 70 | 209 | 104.5 | 85 | M10 | 25 | 45 | 100 |

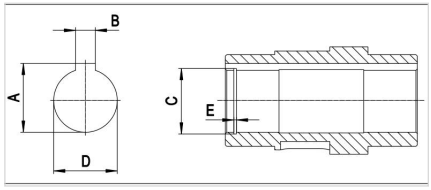



### Reduction bushing with key

Lets you stock reducers with the maximum bore and fit a smaller shaft diameter with a bushing, which cuts stock. The bushing is axially locked with spacers and installation washers and is 5 mm shorter than the hollow shaft. Applies to reducers supplied with the standard hollow shaft diameters.

**RP 165, mm**

| ØD H7 (outer) | ØA H7 (bore options) | C | B | E |
|---|---|---|---|---|
| 75 | 55, 60, 65, 70 | 186 | 58 | 156 |




### Hollow shaft: other bore diameters

Where possible, mount on a shaft the same size as the catalogue bore. Otherwise hollow shafts with other bore diameters can be supplied, up to the maximum bore. Imperial and American bore sizes can also be made, within the maximum.

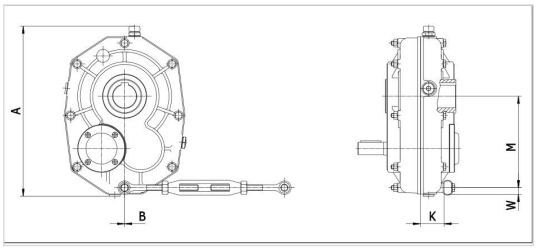
**RP 165, Ø mm H7**

| Catalogue bore | Maximum bore | Special bores available |
|---|---|---|
| 70 | 75 | 60, 65 |

## AVAILABLE HARDWARE

Torque arms, motor mount and backstop

The torque arm anchors the reducer and holds the belts at the correct tension. The standard supply is a torque arm with brackets (dimensions on page 1). Alternatives and accessories for RP 165 are below.

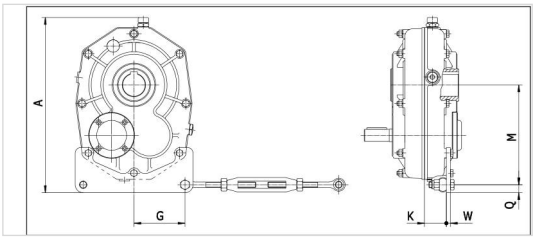



### Torque arm without brackets

Supplied without the mounting brackets, for fixing to your own structure.

**RP 165, mm**

| A | B | M | W | K |
|---|---|---|---|---|
| 498 | 0 | 273.1 | 16 | 68 |

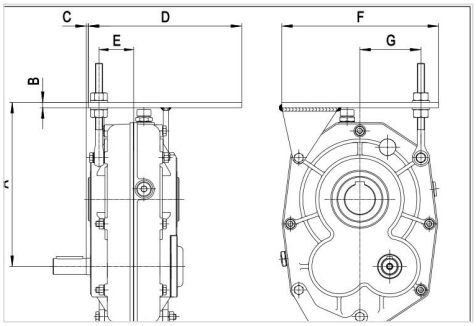



### Torque arm with plate

Torque arm supplied with a mounting plate.

**RP 165, mm**

| A | G | M | Q | W | K |
|---|---|---|---|---|---|
| 534 | 135 | 315 | 21 | 15 | 66 |

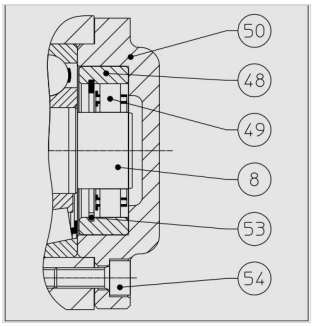



### Motor mount

A compact motor-and-reducer assembly: the belt tension is adjusted at the motor mount torque arm. One motor mount is available for each size, with the maximum motor frame shown.

**RP 165, mm**

| Max. motor (B3) | A | B | C | D | E | F | G | Weight kg |
|---|---|---|---|---|---|---|---|---|
| 180 M | 380 | 14 | 3 | 395 | 76 | 340 | 142.2 | 17.7 |




### Backstop (anti roll-back device)

A freewheel with locking elements, fitted on the input shaft opposite the shaft end, which stops the output shaft turning the wrong way. It can be fitted without removing the reducer from the machine shaft. State the output shaft direction of rotation when ordering, viewed from the side opposite the input shaft. Not available with a double-extended input shaft; on sizes 85 to 165 not with an opposite-side input shaft.

**RP 165**

| Rated input torque M1 | Max. overload (1.7 x M1) |
|---|---|
| 324 Nm | 550.8 Nm |

Check the backstop load capacity: the worst case is low ratios, low service factors and low speeds.

## APPLICATIONS

Where shaft-mounted reducers work hardest

Shaft-mounted reducers drive conveyors, screw conveyors, elevators, mixers and crushers, and are widely used on feeders, screens, stackers and material washers. They suit applications where a belt drive between the motor and the reducer is practical.

- Quick and easy to install: no coupling or base plate needed.
- Rotation in one direction can be prevented with a backstop.
- Output speed can be adjusted by changing the motor and reducer pulleys as well as the gear ratio.
- A range of special executions and accessories.










## OTHER SPECIAL EXECUTIONS

- Special reduction ratios 6.3, 12.5 and 28 on request (not on every size).
- Double-extended input shaft, or input shaft on the opposite side to the catalogue layout.
- Anti-fretting and anti-wear treatment of the hollow shaft.
- Vertical mounting positions VA and VB.

## ORDERING

To select and order, send Q² Machines the required power or torque, output speed, duty and operating position, backstop and torque arm requirements, and the machine shaft diameter. Quote the catalogue designation, for example RP2C150-19.5 ARO with torque arm with brackets. Dimensional drawings in DXF, PDF and STEP formats and spare parts are available on request.